

I E E E S B J I I T

1st STUDENT CONFERENCE RESEARCH FORUM

JAN 30TH – FEB 2ND 2026

JAYPEE INSTITUTE OF
INFORMATION TECHNOLOGY

1st Student Conference Research Forum

Organized by
IEEE Student Branch, IIIT

Proceedings 2026 1st Student Conference of Research Forum

The Student Research Forum (SRF) 2026, organized by the IEEE Student Branch at Jaypee Institute of Information Technology (JIIT), serves as a platform for undergraduate students to explore, present, and discuss their research ideas and innovations.

The forum aims to cultivate a strong research culture among students by encouraging them to investigate real-world problems, develop analytical thinking, and communicate their findings effectively. It provides an opportunity for young researchers to showcase their work, receive constructive feedback, and engage in meaningful discussions with peers and mentors.

SRF 2026 brings together students from diverse domains of engineering and technology, enabling interdisciplinary collaboration and knowledge exchange. Through research presentations, technical discussions, and academic engagement, the forum aspires to inspire students to pursue deeper inquiry and contribute to the advancement of technology and society.

The IEEE Student Branch JIIT remains committed to promoting innovation, creativity, and research excellence among students, and the Student Research Forum is an important step in nurturing the next generation of engineers and researchers.

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Welcome to 2026 1st Edition of the Student Conference: Research Forum

On behalf of Jaypee Institute of Information Technology, Noida, it gives us immense pleasure to welcome you to the 1st Student Conference: Research Forum. This event is being organized by the IEEE Student Branch, Jaypee Institute of Information Technology, Noida, with the objective of providing a dedicated academic platform for students to present research work, exchange ideas, and engage in discussions on emerging and evolving technologies.

In recent years, technological advancements have expanded rapidly across multiple domains, influencing diverse aspects of industry, academia, and society. Fields such as computing systems, electronics, communication technologies, data-driven applications, artificial intelligence, automation, cybersecurity, and other contemporary technological areas continue to redefine problem-solving approaches and innovation practices. Recognizing the interdisciplinary nature of modern technology, this conference has been intentionally designed to be open to a wide range of topics, without restriction to any single specialization.

The conference proceedings consist of student research papers that reflect originality, analytical thinking, and contemporary relevance. These contributions highlight the enthusiasm, creativity, and technical competence of young researchers, while also promoting a culture of scholarly writing and professional presentation. Through paper presentations and academic interactions, participants gain valuable exposure to research methodologies, interdisciplinary perspectives, and emerging challenges in technology.

This initiative represents a step toward strengthening research-oriented learning and fostering innovation-driven thinking among students. We sincerely hope that the conference serves as a meaningful academic experience and inspires participants to pursue further research, innovation, and development activities.

We extend a warm welcome to all participants and wish the conference every success.

The Research Forum included academic sessions conducted by faculty members who guided participants on research writing, abstract development, and poster-based presentation.

Keynote Speaker

Dr. Alka Singhal

(Assistant Professor (Senior Grade),JIIT)

Dr. Nidhi Tewari

(Assistant Professor (Senior Grade),JIIT)

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Energy–Accuracy Trade-off Analysis of MobileNet-V2 for Edge AI Inference

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Abstract—Edge artificial intelligence deployments are constrained by limited energy availability and thermal budgets, making accuracy-centric evaluation insufficient for real-world systems. This paper presents a comparative meta-analysis of MobileNet-V2 inference under FP32, FP16, and INT8 numerical precision on Jetson-class edge hardware. Benchmark-reported accuracy, inference latency, and power measurements were collected from publicly available vendor documentation and prior studies. Energy per inference was computed by synthesizing reported power and latency metrics, enabling a cross-metric evaluation that is often absent in isolated benchmark reports. The results show that FP16 inference significantly reduces energy consumption while preserving accuracy, whereas INT8 inference achieves the lowest energy per inference with a small accuracy reduction. Pareto frontier analysis highlights the trade-offs between accuracy and energy efficiency and the dominance of GPU-based inference over CPU execution.

Keywords— Edge AI, MobileNet-V2, Quantization, Energy per Inference, Power Consumption, Pareto Frontier

I. INTRODUCTION

Deep neural networks are increasingly deployed on embedded and edge platforms for applications such as computer vision, robotics, and intelligent sensing. While model development is often guided by accuracy benchmarks, practical edge deployments must operate under strict constraints on energy consumption, thermal dissipation, and real-time latency.

Most prior works report either accuracy or latency in isolation, with limited emphasis on energy efficiency as a first-class metric. This paper addresses this gap by performing a cross-metric synthesis of reported benchmark results, enabling a unified evaluation of accuracy, latency, power, and energy per inference for MobileNet-V2 under different numerical precision modes.

II. METHODOLOGY

A. Evaluation Framework

This study follows a comparative meta-analysis approach. Reported benchmark measurements for MobileNet-V2 inference were collected from NVIDIA Jetson benchmarking documentation, MLPerf Inference benchmark reports, and peer-reviewed studies focusing on edge AI optimization. The evaluation framework targets Jetson-class embedded platforms and GPU-accelerated inference paths. No direct hardware measurements were performed as part of this work.

B. Energy Computation

Energy per inference was computed using reported average power and inference latency values as:

$$E_{\text{inf}} = P_{\text{avg}} \times T_{\text{inf}} \quad (1)$$

where P_{avg} denotes the average power consumption (W) and T_{inf} denotes the inference latency (s). Energy values are reported in millijoules (mJ). All units were normalized to ensure consistency across sources.

III. RESULTS

A. Precision-Level Comparison

TABLE I
PRECISION VS ACCURACY, LATENCY, POWER, AND ENERGY FOR MOBILENET-V2

Precision	Accuracy (%)	Latency (ms)	Power (W)	Energy (mJ)
FP32	71.8	54.2	0.26	14.10
FP16	71.8	30.0	0.28	8.46
INT8	71.2	18.0	0.31	5.64

B. CPU vs GPU Execution

TABLE II
CPU VS GPU INFERENCE PERFORMANCE FOR MOBILENET-V2

Execution Mode	Precision	Accuracy (%)	Latency (ms)	Power (W)	Energy (mJ)
CPU-only	FP32	71.8	95.0	0.26	24.7
GPU	FP32	71.8	54.2	0.26	14.1
GPU	FP16	71.8	30.0	0.28	8.46
GPU	INT8	71.2	18.0	0.31	5.64

C. Observed Trends

- FP16 inference reduces energy per inference by approximately **40%** relative to FP32 while preserving identical Top-1 accuracy.
- INT8 inference achieves the lowest energy consumption, reducing energy per inference by approximately **60%** compared to FP32 with less than **1%** accuracy degradation.
- Inference latency decreases monotonically with reduced numerical precision, indicating improved utilization of GPU accelerators.

- CPU-based inference is dominated by GPU execution across both latency and energy metrics.

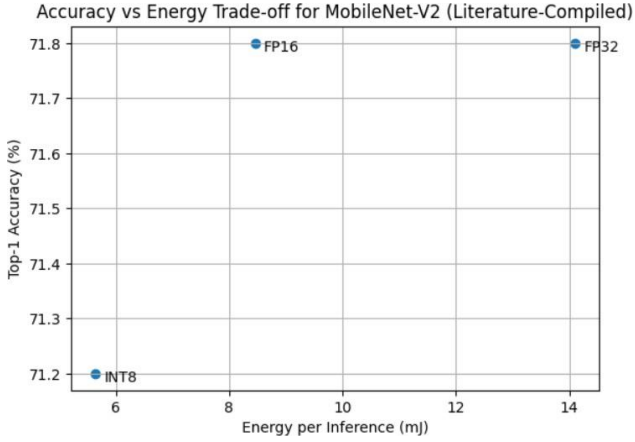


Fig. 1. Accuracy versus energy per inference Pareto frontier for MobileNet-V2 under different numerical precision modes.

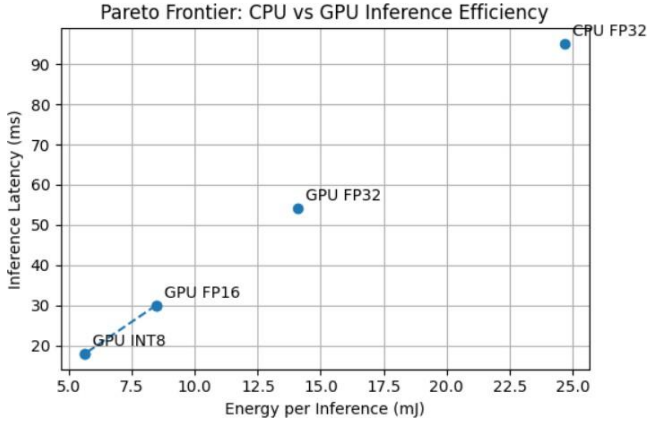


Fig. 2. Pareto frontier comparing CPU and GPU inference efficiency for MobileNet-V2

IV. CONCLUSION

This paper presented a comparative meta-analysis of MobileNet-V2 inference on Jetson-class edge hardware, with a specific focus on energy per inference as a deployment-relevant metric. By synthesizing benchmark-reported accuracy, latency, and power measurements from multiple publicly available sources, the study quantified the trade-offs introduced by numerical precision and execution hardware.

The analysis shows that FP16 inference provides a balanced operating point, achieving approximately **40%** reduction in energy per inference relative to FP32 without any observable loss in Top-1 accuracy. INT8 inference further improves energy efficiency, reducing energy per inference by approximately **60%** while maintaining accuracy within **1%** of the FP32 baseline when appropriate quantization techniques are applied. GPU-based inference consistently outperforms CPU execution, exhibiting lower latency and reduced energy consumption across all evaluated precision modes.

The Pareto frontier analysis highlights that improvements in energy efficiency are closely coupled with reduced numerical precision and hardware acceleration. While the analysis relies on benchmark-reported measurements collected under varying experimental conditions, the cross-metric synthesis provides

insights that are often absent in individual benchmark studies. All numerical values reported in this work are derived from publicly available benchmark results and datasets and are used solely for comparative analysis. These findings reinforce the importance of energy-centric evaluation when selecting models and precision modes for edge AI deployments.

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